

$$\oint \vec{V} \cdot \vec{F} = 0 \Rightarrow \left[\iint_{S_2} \vec{F} \cdot \vec{n}_2 \cdot d\vec{S} \Rightarrow \iint_{S_1} \vec{F} \cdot \vec{n}_1 \cdot d\vec{S} \right]$$

Recitation.

Q1 $\langle 1, 2, -1 \rangle$

$$F_n \quad a \langle x - x_0 \rangle + b \langle y - y_0 \rangle + c \langle z - z_0 \rangle = 0$$

$$\langle x - 0 \rangle + 2 \langle$$

simple plane =

$$x + 2y - z = 0$$

$$z = x + 2y$$

let $x = u$
 $y = v$
 $z = u + 2v$

$$a \langle u, v, u + 2v \rangle$$

Q21- $r(\rho, \phi) = (a \sinh \phi \cos \rho, a \sinh \phi \sin \rho, a \cosh \phi)$

Q140- $f(x,y) = 4-x-y$

$$f_x = 0 - 1 = -1$$

$$f_y = -1$$

$$\sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + 1^2 + 1^2} = \sqrt{3}$$

$$x+y+z = x+y+(4-x-y) = 4$$

So,

$$4\sqrt{3}, \text{ Area of base of rectangle} = 2 \cdot 1 = 2$$

$$\iint_S (x+y+z) \cdot dS = 4\sqrt{3} \cdot 2 = 8\sqrt{3}$$

Q141- $\iint_S z \cdot dS$ if $z = 4 - x^2 - y^2$ above $z=0$

$$f_x = -2x$$

$$f_y = -2y$$

$$\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2} \cdot r \, dr$$

$$A = 2\pi \int_0^2 r \cdot \sqrt{1 + 4r^2} \cdot dr = \left[\frac{\pi}{6} (17^{3/2} - 1) \right]$$

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Q71- $r^2 = 4$

$r = 2$

$$\text{Area} = 2\pi(2)(3) = \boxed{12\pi}$$

or

$$r(\theta, z) = (r \cos \theta, r \sin \theta, z)$$

$$r(\theta) = (-r \sin \theta, r \cos \theta, 0)$$

$$r(z) = (0, 0, 1)$$

$$r_\theta \times r_z = \begin{vmatrix} i & j & k \\ -r \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = i(r \cos \theta) - j(-r \sin \theta) + k$$

$$= r \sin \theta j + r \cos \theta i$$

$$= \sqrt{r^2 \sin^2 \theta + r^2 \cos^2 \theta} = \boxed{r} \rightarrow 2$$

$$\int_0^3 \int_0^{2\pi} (r_\theta \times r_z) \cdot r \, d\theta \, dz$$

$$\boxed{\int_0^3 \int_0^{2\pi} 2 \cdot dr \cdot d\theta}$$

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